FUEL ANALYSIS REPORT

REPORT ID: S260854862	CUSTOMER: HUAYU SHIP MANAGEMENT LIMITED	SHIP(IMO NO): XINSHUN(9252199)	BUNKER PORT: GENT

CONFORMANCE

The fuel sample tested conforms to Table 1 of ISO 8217:2005 specifications for grade MGO - DMA LS

*Conformance to table 1 or table 2 of ISO 8217 does not imply conformance to other sections of ISO 8217 such as para 5.

BUNKER INFORMATION		BDN/CLIENT INFORMATION		SAMPLE DETAILS	
BUNKER DATE	01-Aug-2026	GRADE	DMA LS	SENT FROM	
BUNKER PORT	GENT	DENSITY@15°C kg/m³	886.0	SENT DATE	5-Aug-2026
SUPPLIER	NORTHSTAR BUNKER	VISCOSITY@40°C cSt	5.7	REPORT DATE	25-Aug-2026
BARGE	BREITLING	SULFUR %m/m	0.010	TEST DATE	24-Aug-2026
SAMPLING POINT	MANIFOLD	WATER %v/v	-	RETENTION DATE	24-Sep-2026
SAMPLING METHOD	DRIP	QUANTITY (MT)	99.496	PO NUMBER	-

SEAL INFORMATION

LAB SEAL	MARPOL SEAL	SHIP SEAL	BARGE SEAL
D593342(Sealed)	D593279		

RECOMMENDED TEMPERATURES

A minimum viscosity has to be maintained at engine inlet and for practical purpose some engine makers recommend this to be not less than 2cSt. Based on tested viscosity, this oil should not be heated above 78°C temperature in order to meet the requirement.

CALCULATIONS OF QUANTITY SUPPLIED BASED ON LAB TESTS

	UNITS	BUNKER INFORMATION	TESTED RESULTS
Density @15°C	kg/m³	886.000	885.800
Water	%v/v	-	0.000
Quantity supplied as per ship	MT	99.496	
Quantity shortage due to density	MT	0.022	
Quantity shortage due to water	MT	-	
Total Quantity Shortage	MT	0.022	

*Above figures do not account for ISO 4259 criteria given in ISO 8217 Annex F. (This is not applicable for delivery made by mass flowmeter)

TEST DESCRIPTION	UNITS	RESULTS	ISO 8217 LIMITS	METHOD
Density @15°C	kg/m³	885.8	Max 890.0	ISO 12185
Viscosity @ 40°C	cSt	5.6	Min 1.5 Max 6.0	ISO 3104
Flash Point	°C	> 70.0	Min 60.0	ISO 3679
Pour Point	°C	0	Max(Summer) 0 Max(Winter) -6	ISO 3016
Micro Carbon Residue	% m/m	0.01	Max 0.30	ISO 10370
Ash	% m/m	0.003	Max 0.010	Calculated
Water	% v/v	< 0.01	-	ASTM D6304
Sulfur	% m/m	0.097	Max 0.10	ISO 8754
Vanadium	mg/kg	< 1	-	VLS-LM8
Al+Si (Catfines)	mg/kg	2	-	VLS-LM8
Sodium	mg/kg	< 1	-	VLS-LM8
Aluminum	mg/kg	< 1	-	VLS-LM8
Silicon	mg/kg	< 1	-	VLS-LM8
Iron	mg/kg	< 1	-	VLS-LM8
Nickel	mg/kg	< 1	-	VLS-LM8
Calcium+	mg/kg	< 1	-	VLS-LM8
Phosphorous+	mg/kg	< 1	-	VLS-LM8
Zinc+	mg/kg	< 1	-	VLS-LM8
Potassium	mg/kg	< 1	-	VLS-LM8
Lead	mg/kg	< 1	-	VLS-LM8
Magnesium	mg/kg	< 1	-	VLS-LM8
API Gravity	-	29.02	-	-
Net Specific Energy	MJ/kg	43.050	-	Calculated
Gross Specific Energy	MJ/kg	45.850	-	Calculated
Visual Appearance		Clear & Bright	-	LM**

LM** - LAB METHOD

Calcium > 30, Zinc > 15 and Phosphorus > 15 (ISO 8217:2005 standard)

VESSEL PAST BUNKERING HISTORY DATA					
		S260854862			
BUNKER DATE		01-Aug-2026			
BUNKER PORT		GENT			
QUANTITY (MT)		99.496			
TEST DESCRIPTION	UNITS				
Density @15°C	kg/m³	885.8			
Viscosity @ 40°C	cSt	5.6			
Flash Point	°C	> 70.0			
Pour Point	°C	0			
Micro Carbon Residue	% m/m	0.01			
Ash	% m/m	0.003			
Water	% v/v	< 0.01			
Sulfur	% m/m	0.097			
Vanadium	mg/kg	< 1			
Al+Si (Catfines)	mg/kg	2			
Sodium	mg/kg	< 1			
Aluminum	mg/kg	< 1			
Silicon	mg/kg	< 1			
Iron	mg/kg	< 1			
Nickel	mg/kg	< 1			
Calcium	mg/kg	< 1			
Phosphorous	mg/kg	< 1			
Zinc	mg/kg	< 1			
Potassium	mg/kg	< 1			
Lead	mg/kg	< 1			
Magnesium	mg/kg	< 1			
API Gravity		29.02			
Net Specific Energy	MJ/kg	43.050			
Gross Specific Energy	MJ/kg	45.850			
Visual Appearance		Clear & Bright			

On behalf of Viswa Lab

Signed:

S.Vkl

VIGNESH KUMAR S

Date: 25-Aug-2026

- *This report shall not be reproduced except in full, without the written approval of the laboratory.*
- *Precision parameters apply to the test results. The above test results relate only to the item tested.*
- *Viswa Lab assumes no responsibility and shall not be liable to any person for any loss, damage or expense caused by reliance on the information or advice in this document or however provided, unless that person has signed a contract with Viswa Lab for the provision of this information •or advice and in that case any responsibility or liability is exclusively on the terms and conditions set out in that contract.*
- *EFN* (Engine Friendliness Number) - EFN will not be reported for any offspec or interim reports. For further details about Engine Friendliness Number refer our Website.*
- *Lab Name and Address : Viswa Lab Singapore, 80 Tagore Lane, Singapore 787501*

UNIQUE REPORT IDENTIFIER :###VLS BThzx0ZYMZefmQ8W6Njnp1clZowh4eZXzr368P+nNs8= VLS###